DPRL Assignment 2

- Implement the following exercise in python using standard functions and packages (no MDC packages)
- Consider a system that slowly deteriorates over time:
  - When it is new it has a failure probability of 0.1
  - the probability of failure increases every time unit linearly with 0.01
  - Replacement costs are 1, after replacement the part is new

a) What is an appropriate state space? Argue why this satisfies the Markov property

b) Compute the stationary distribution and use this to find the long-run average replacement costs

c) Simulate the system for a long period and verify that you get (approximately) the same answer

d) Solve the average-cost Poisson equation

e) Preventive replacement is possible at cost 0.6. What is the average optimal policy? Solve it using:

- policy iteration
- value iteration

How and what to submit

- Report (.pdf) of max 2 A4 pages plus appendix with relevant figures/tables/screenshots OR max 1000 words
- Separate Python code file (.py) or jupyter notebook
- Implement the algorithm in an efficient way, it should run very fast
- Report should include the solution method. Mathematically describe the method that you coded, including implementation choices and initialization and stopping criteria of the method
- Comment on your findings, are they as expected?
- Grading:
  - 1 + a + 2 b + c + 2 d + 2 e (PI) + 1 e (VI)
  - description and implementation are both graded